

Lesson 13: Systems of Differential Equations - Modeling & Elimination

Professor Dave Explains Differential Equations Master Series · Lesson 13

Video: Systems of Differential Equations Part 1: Modeling and Elimination **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that in engineering, variables rarely change in isolation. In interconnected chemical reactors, cooling loops, or coupled predator-prey ecosystems, the rate of change of substance x depends on y , and the rate of change of y depends on x . By defining the differential operator $D = d/dt$, we treat derivatives as algebraic multipliers, allowing us to systematically eliminate one variable and collapse the coupled system into a single higher-order ODE.

2. Core Theoretical Framework & Rigorous Formulations

1. Interconnected Fluid Mixing Model

$$\begin{aligned}\frac{dx_1}{dt} &= \text{Rate In}_1 - \text{Rate Out}_1 = c_{in}R_{in} + \frac{x_2}{V_2}R_{21} - \frac{x_1}{V_1}(R_{12} + R_{1,out}) \\ \frac{dx_2}{dt} &= \text{Rate In}_2 - \text{Rate Out}_2 = \frac{x_1}{V_1}R_{12} - \frac{x_2}{V_2}(R_{21} + R_{2,out})\end{aligned}$$

2. Differential Operator Elimination Method

Let $D = \frac{d}{dt}$. Write the linear system as:

$$L_1x + L_2y = g_1(t)$$

$$L_3x + L_4y = g_2(t)$$

Operate to eliminate y : $(L_1L_4 - L_2L_3)x = L_4g_1(t) - L_2g_2(t)$. Solve the scalar ODE for $x(t)$, then substitute back to find $y(t)$ without introducing redundant constants.

3. Fully Solved Whiteboard Problems

Exam Problem: Solving a Coupled 2x2 System by Systematic Elimination

Problem Statement: Solve the coupled system $\begin{cases} x' = 2x - y \\ y' = x \end{cases}$ with $x(0) = 1, y(0) = 0$.

Step 1: Write in Differential Operator Notation

$(D - 2)x + y = 0$ (Eq 1) and $-x + Dy = 0$ (Eq 2).

Step 2: Eliminate y by Algebraic Combination

Operate on Eq 1 with D : $D(D - 2)x + Dy = 0$. Subtract Eq 2: $D(D - 2)x - (-x) = 0 \implies (D^2 - 2D + 1)x = 0$.

Step 3: Solve Scalar Second-Order ODE for $x(t)$

Characteristic equation: $r^2 - 2r + 1 = (r - 1)^2 = 0 \implies r = 1$ (repeated). $x(t) = c_1e^t + c_2te^t$.

Step 4: Express $y(t)$ Directly from System

From Eq 1: $y(t) = 2x - x'$. Compute $x'(t) = c_1e^t + c_2(e^t + te^t) = (c_1 + c_2)e^t + c_2te^t$. Therefore: $y(t) = 2(c_1e^t + c_2te^t) - [(c_1 + c_2)e^t + c_2te^t] = (c_1 - c_2)e^t + c_2te^t$.

Step 5: Enforce Initial Conditions

$x(0) = c_1 = 1$. $y(0) = c_1 - c_2 = 0 \implies c_2 = c_1 = 1$. Final solution: $x(t) = (1 + t)e^t$, $y(t) = te^t$.

⚠ Common Exam Pitfall & Trapdoor Avoidance

Never integrate a separate second-order ODE for $y(t)$ independently—doing so introduces two new constants (c_3, c_4) that must be laboriously eliminated. Always deduce $y(t)$ directly from the first-order coupling equation.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

Operator D converts systems into single higher-order ODEs. Preserve constant dependency by expressing the second variable directly.

- Video reference: [Systems of Differential Equations Part 1: Modeling and Elimination](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.